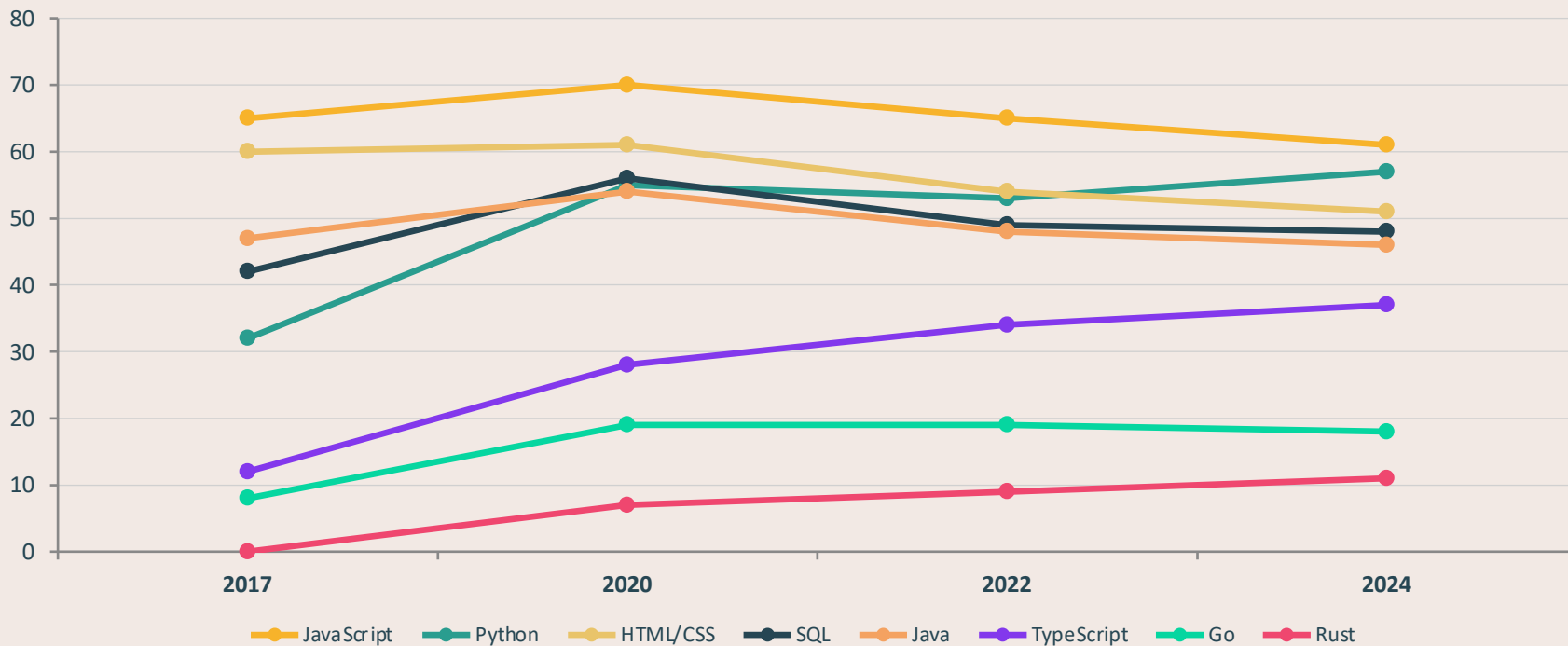


State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

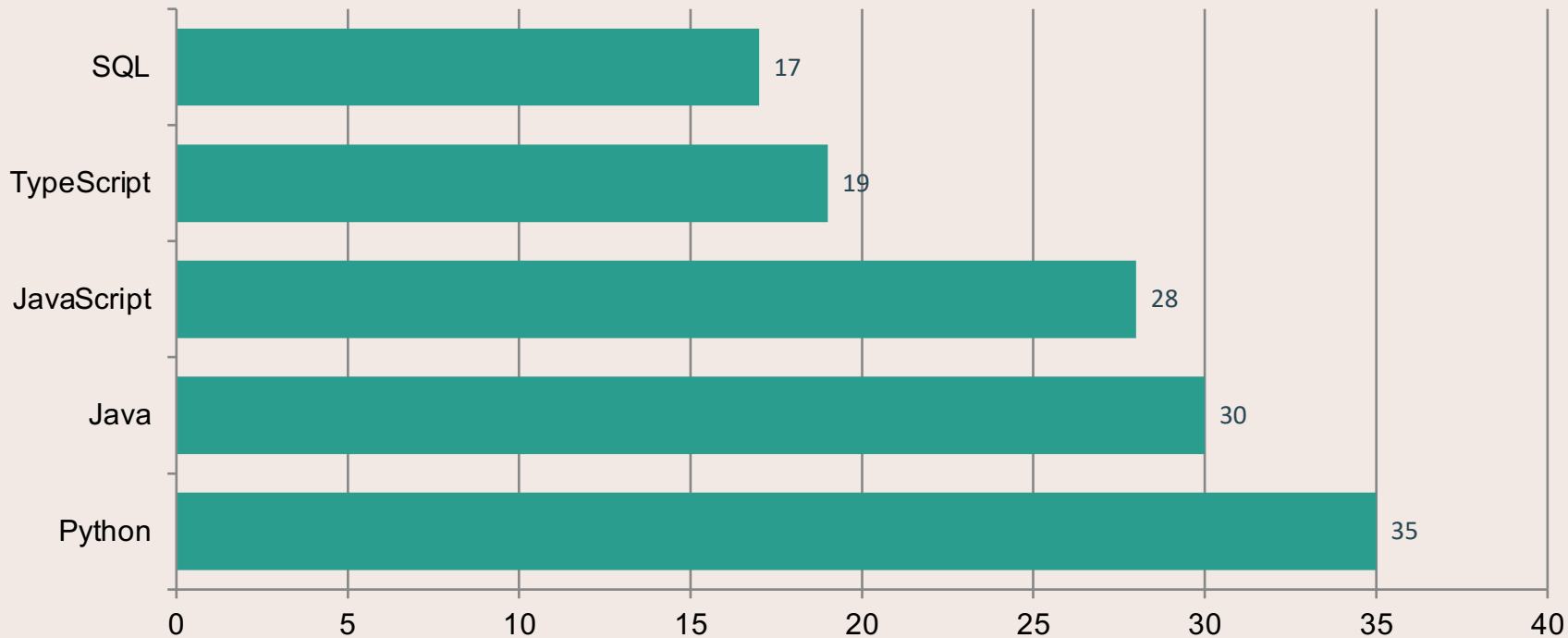
Insights from JetBrains' survey of 23,262 developers worldwide

Python Closes the Gap: 8 Years of Language Usage Shifts



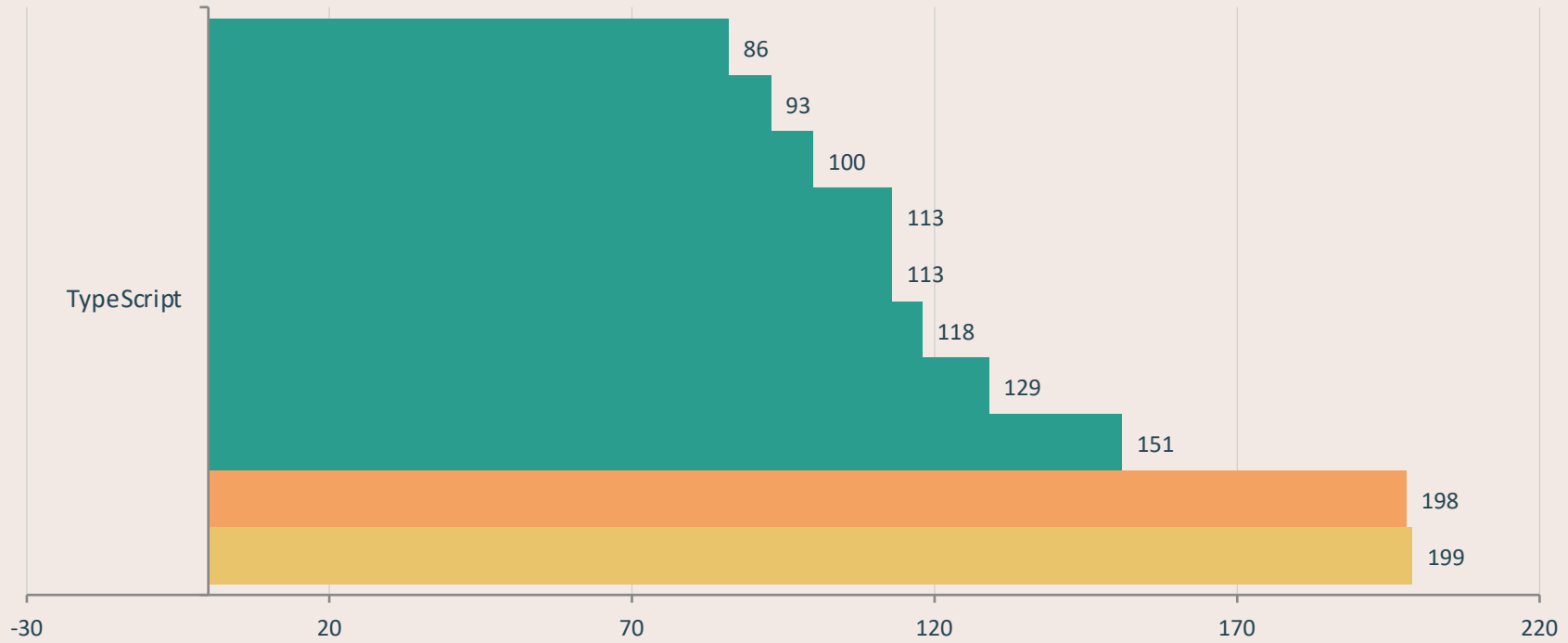
Source: JetBrains State of Developer Ecosystem 2024

Python Is Now the #1 Primary Language at 35%



Note: Adoption intent is below 6% for all languages

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Language dominance:



Python



TypeScript



Java



C++

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML teams — it is the undisputed leader

2

TypeScript is the future of frontend and full-stack

3

Rust and Go are the growth plays for infrastructure and systems

4

PHP and Ruby hiring pools are shrinking

5

Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy